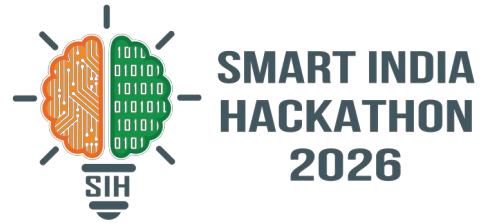


SMART INDIA HACKATHON 2026



IDEA SUBMISSION · SIH26237 · SOFTWARE

NISHAN-PQ

निशान = "mark"

Every decrypted copy leaves a verifiable mark.

Post-quantum document distribution in which every authorised decryption creates a uniquely marked, recipient-signed, quorum-logged copy — so a leaked copy leads back to its release session.

PROBLEM STATEMENT ID	SIH26237
PROBLEM STATEMENT TITLE	Cryptographic Attribution and Immutable Decryption Provenance for Multi-Recipient Encrypted Document Distribution
ORGANISATION	Ministry of Defence · Indian Navy (WESEE)
THEME	Blockchain & Cybersecurity
PS CATEGORY	Software
TEAM ID	152425
TEAM NAME (REGISTERED ON PORTAL)	Aevrix
COLLEGE NAME & LOCATION	Global Academy of Technology, Bengaluru



github.com/ashebbbar-dev/SIH

open source · offline · reproducible

Working prototype · fresh run, 28 Sep 2026

59/59

automated tests passed

12/12

security safeguard scenarios

25/25

attack-benchmark checks met

0

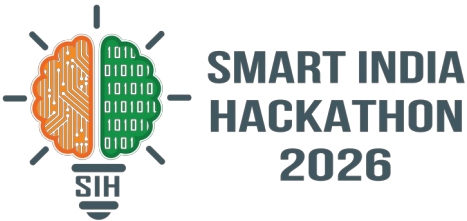
false accusations in 175 attack cases

ML-KEM-768 + ML-DSA-65

real NIST post-quantum crypto (FIPS 203 / 204)



IDEA TITLE: NISHAN-PQ



Every decrypted copy leaves a verifiable mark

post-quantum · offline · two-channel forensic mark · quorum-signed provenance

THE PROBLEM · ROOT CAUSE

Same file. 1,000 readers. One leak.

Whose copy was it?



- Encrypt-once: every reader decrypts identical bytes
- Access logs show who opened it — not whose copy leaked
- Whoever runs the log can quietly rewrite it
- RSA / ECC are quantum-vulnerable (NIST: deprecate after 2030)

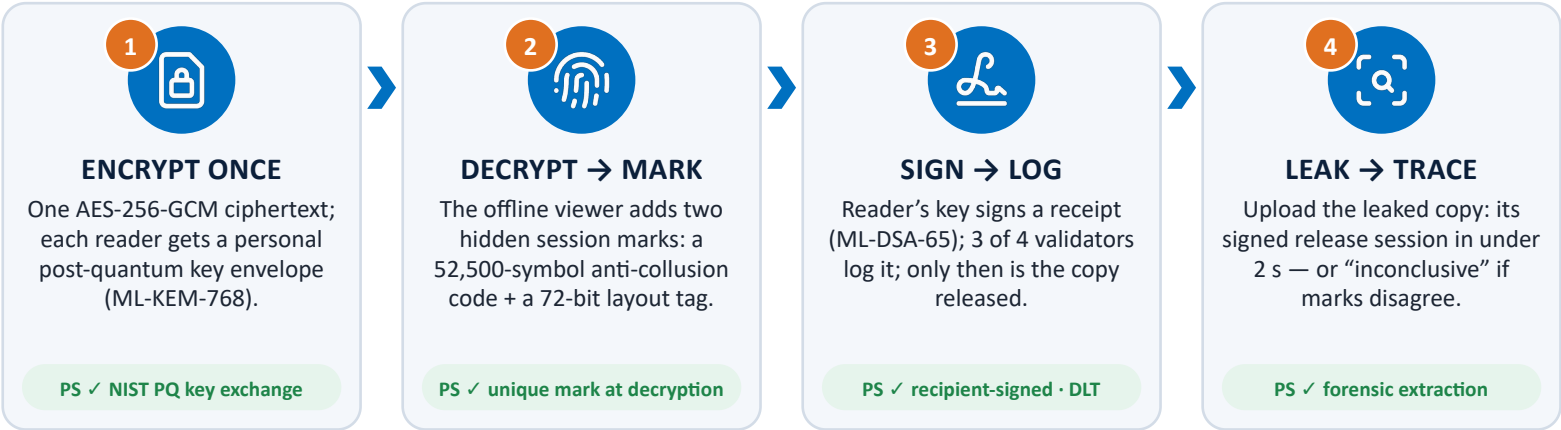
Real risk: in 2016, 22,400 pages of Scorpène submarine data (Indian Navy programme) leaked to the press [14].

INNOVATION AND UNIQUENESS

Two independent marks must agree — or NISHAN refuses to accuse.

- Pasted, stripped or conflicting marks ⇒ “inconclusive”, never a wrong name
- Commit-before-release: no signed, quorum-logged receipt ⇒ no copy
- Post-quantum end to end (NIST FIPS 203 / 204), fully air-gapped

PROPOSED SOLUTION *detailed explanation — the user journey*



HOW IT ADDRESSES THE PROBLEM: all 6 SIH26237 requirements implemented & tested — PQ key exchange · mark at decryption · recipient's signature · immutable log · forensic extraction · air-gapped

TARGET BENEFICIARIES

- Document controllers**
Indian Navy / WESEE, MoD cells
- Leak investigators**
security & vigilance officers
- Honest recipients**
protected from being framed
- Auditors & command**
a history no one can quietly edit

IMPACT METRICS *measured in prototype tests*

–99.9%

suspect pool: 1,000 possible sources
→ 1 signed session

0

false accusations in 175 attack cases

< 2 s

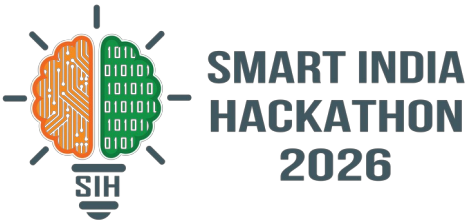
leak-to-session trace, all 1,000 rows scored

8 / 8

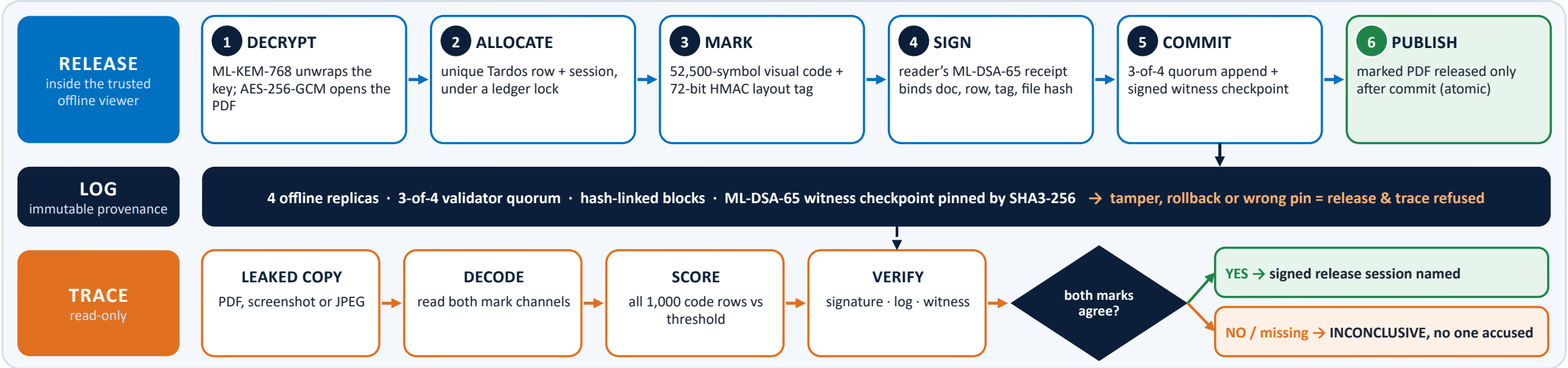
tamper, forgery & rollback attacks caught



TECHNICAL APPROACH



METHODOLOGY AND PROCESS FOR IMPLEMENTATION architecture of the working prototype (numbered = implementation steps)



WORKING PROTOTYPE real output, 28 Sep 2026

RESTRICTED // EXERCISE

Operations Readiness

Distribution: ALPHA CELL

This synthetic document is generated on
The sender encrypts it once. Each autho

Source PDF (synthetic)

RESTRICTED // EXERCISE

Operations Readiness

Distribution: ALPHA CELL

This synthetic document is generated on
The sender encrypts it once. Each autho

Marked copy (Alice)

Reads exactly like the source · text 100 % preserved & searchable ·
41.9 dB PSNR · a different code for every session

TECHNOLOGIES TO BE USED

Python 3 · OpenSSL 3.6 (ML-KEM-768, ML-DSA-65) · AES-256-GCM ·
SHA3-256 · PyMuPDF · NumPy / OpenCV · offline browser UI

DATA SOURCES

Controller's source PDFs · enrolled reader keys · per-document
codebook secret · leaked copy from the investigator. No external APIs;
tests use synthetic documents.

PLATFORM & DEPLOYMENT

Air-gapped LAN: viewer kiosk per reader, 4 validator nodes + 1 witness,
investigator console. Linux PCs; no GPU, no Internet.

TRACE RESULTS real runs, 28 Sep 2026

```
decision corroborated_channels
recipient alice · session aee9e454...
score 17,252 (threshold 2,100)
layout 72-bit HMAC tag authenticated
receipt ML-DSA-65 valid · witness ok
pasted mark on re-rendered copy → abstain
```

1.9 s

release per copy

1.0 s

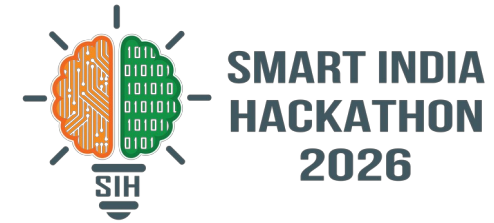
trace per leak

0.46 s

score 1,000 rows



FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA *it already runs: fresh results on an offline laptop, 28 Sep 2026*

TESTED TODAY — ALL GREEN

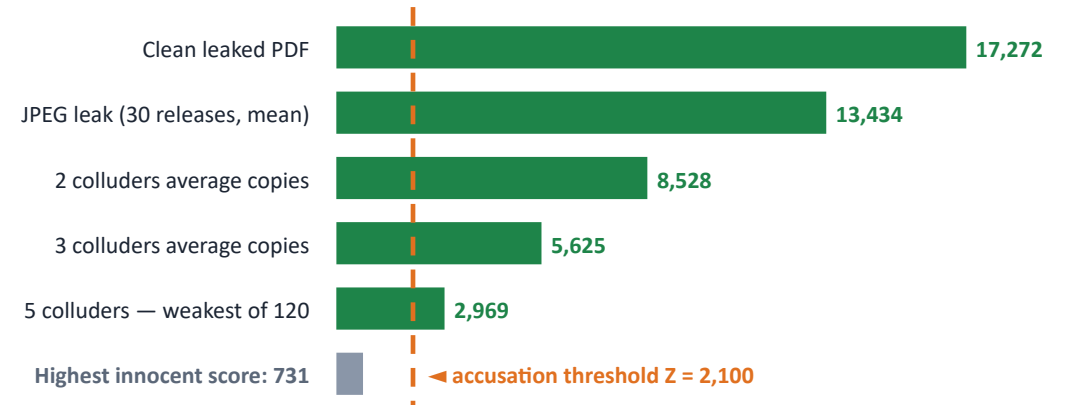
✓ Unit & integration tests 12 test modules · 54 s		59/59
✓ Release-safeguard scenarios concurrency · rollback · wrong pin · failure · 24 s		12/12
✓ Attack benchmark PDF edits · JPEG · crop · rotate · collusion · 29 s		25/25
✓ Collusion study 30 codebooks × 4 strategies, 5 colluders · 29 s		120/120
✓ End-to-end real-PQC releases decrypt → mark → sign → log → JPEG leak → trace · 89 s		30/30

OpenSSL 3.6.4 · ML-KEM-768 ✓ · ML-DSA-65 ✓

0 false accusations in 175 attack cases

GUILTY STAND OUT, INNOCENTS STAY FAR BELOW

Tardos accusation score by leak type (fresh runs; higher = stronger evidence)



Across 145 attack cases every innocent row stayed $\geq 2.8\times$ below the threshold.

CHALLENGES & RISKS → STRATEGIES FOR OVERCOMING THEM

Admin rewrites history	3-of-4 quorum + pinned ML-DSA witness; rollback ⇒ release refused	TESTED
Mark pasted or stripped	two channels must agree, else abstain — never a wrong name	TESTED
Future quantum decryption	ML-KEM-768 + ML-DSA-65 today; no RSA/ECC fallback (against harvest-now-decrypt-later)	BUILT
Plaintext grabbed before marking	marking inside the trusted viewer → signed kiosk + hardware keys	ROADMAP
Leak by print or phone photo	signed compact token (13 + 63 tests pass); 24 research rounds on real prints	RESEARCH

36-HOUR FINALE SCOPE · MVP

Live MVP: encrypt once → 3 readers → leak one copy → traced < 2 s → tamper the log → release refused

Build in 36 h: ① validators + witness on 5 separate machines (isolated LAN) ② multi-page batch release ③ signed investigator report

Team of 6: security architect · PQC & identity · watermark research · ledger · UX & demo · evaluation

Resources in hand: open-source stack, laptop-class hardware, offline; nothing to buy or license for the finale

VIABILITY

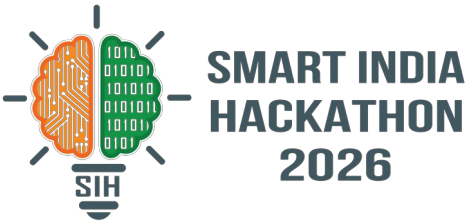
↗ **Market:** every multi-recipient restricted-document flow: MoD, services, DRDO, PSUs, ministries, banks; insider risk costs US\$19.5 M per organisation per year [12]

₹ **Budget (indicative):** ₹0 licences (open source); pilot ≈ ₹3 lakh for 5 commodity nodes; runs on existing PCs

📦 **Scalability:** 10× more readers → only ~14 % longer code; validators scale out



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE metrics measured on the prototype

Document controller
Every copy is signed & logged before it leaves
every released file matched its signed receipt

Leak investigator
Leaked copy → its release session, fully offline
1,000 → 1 session in < 2 s

Honest officer
Can't be framed by a pasted or stripped mark
0 false accusations in 175 attacks

Command & auditors
A history no admin can quietly rewrite
8 / 8 tamper & rollback attacks caught

COMPETITIVE ADVANTAGE how NISHAN-PQ compares with conventional approaches

✓ yes − partial ⊗ no

typical capability per approach · NISHAN-PQ column measured in our tests

	Access logs / DRM	Visible name stamp	Single hidden watermark	Watermark + blockchain	NISHAN-PQ
Names the release session of a leaked copy	⊗	−	✓	✓	✓
Resists colluders averaging their copies	⊗	⊗	−	−	✓
Tamper-evident, quorum-signed history	⊗	⊗	⊗	✓	✓
Refuses to accuse on conflicting evidence	⊗	⊗	⊗	⊗	✓
Post-quantum crypto (NIST FIPS 203 / 204)	⊗	⊗	⊗	⊗	✓
Air-gapped, text stays searchable	✓	✓	−	⊗	✓

BENEFITS OF THE SOLUTION social · economic · environmental · sustainability

Social & security: deters insider leaks; protects honest personnel from suspicion

Economic: faster, evidence-based probes; avg Indian breach ₹22 crore [11]; ₹0 licences

Environmental: controlled digital release instead of numbered paper copies; no GPU / cloud

Sustainable: quantum-safe beyond the 2030–35 RSA/ECC phase-out [8]; open standards

ROADMAP

SEP 2026 ✓

DEC 2026

H1 2027

H2 2027

Working prototype: 59/59 tests, 12/12 safeguards, 25/25 attack checks

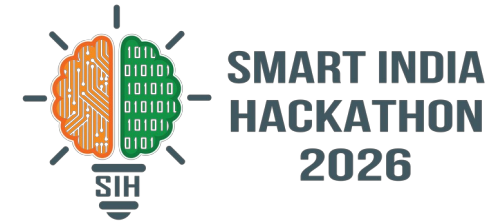
Finale: multi-node air-gapped deployment, multi-page PDFs, signed report

Pilot with a document cell: hardware-backed keys, signed kiosk viewer

Print & photo channel (signed compact token) + independent security audit



RESEARCH AND REFERENCES



DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

RESEARCH DEPTH *git history 15–28 Sep 2026 + evidence folder*

291

commits across 61 branches

24

print & phone-photo research rounds (R1–R24)

887

test functions on research branches

35

dated research & decision records

62

evidence records: JSON results, reviews, diffs

175

attack cases run — 0 false accusations

RELATED WORK *what is established → what NISHAN-PQ adds*

Collusion-secure codes

Boneh–Shaw [2], Tardos [1]



full 52,500-symbol row inside a live, searchable PDF; all 1,000 rows scored in 0.46 s

Text-layout marking

Brassil et al. [3]



72-bit HMAC session tag bound to the same signed release

Watermark + blockchain

T-Tracer [6]



PQ signatures, commit-before-release, witness catches full rollback

PQ permissioned ledgers

PQFabric [4]



lightweight 3-of-4 offline quorum log for air-gapped networks

Insider-threat detection

deep-learning review [5]



adds per-copy cryptographic evidence after a leak happens

IP POTENTIAL: the conflict-aware two-channel decision rule (attribute only when both marks agree, else abstain) bound to one PQ-signed, quorum-committed release, plus the commit-before-release gate. Candidate claims, subject to a formal patent search.

REFERENCES *each checked to exist, 28 Sep 2026*

ACADEMIC — IEEE · ELSEVIER · ACM

- [1] G. Tardos, “Optimal probabilistic fingerprint codes,” J. ACM, vol. 55, no. 2, 2008.
- [2] D. Boneh, J. Shaw, “Collusion-secure fingerprinting for digital data,” IEEE Trans. Inf. Theory, 44(5), 1998.
- [3] J. Brassil et al., “Electronic marking and identification techniques to discourage document copying,” IEEE JSAC, 1995.
- [4] A. Holcomb et al., “PQFabric: a permissioned blockchain secure from both classical and quantum attacks,” IEEE ICBC, 2021.
- [5] S. Yuan, X. Wu, “Deep learning for insider threat detection: review, challenges and opportunities,” Computers & Security (Elsevier), 104, 2021.
- [6] L. Wang et al., “T-Tracer: a blockchain-aided symbol mapping watermarking scheme for traitor tracing,” ACM BSCI, 2022.

STANDARDS & TECHNOLOGY DOCUMENTATION

- [7] NIST FIPS 203 (ML-KEM) and FIPS 204 (ML-DSA), Aug 2024 — csrc.nist.gov/pubs/fips/203/final
- [8] NIST IR 8547 (draft), Transition to Post-Quantum Cryptography Standards, 2024: RSA/ECC deprecated after 2030.
- [9] OpenSSL 3.5 LTS (Apr 2025): native ML-KEM / ML-DSA; prototype runs OpenSSL 3.6.4.
- [10] SIH26237 problem statement, Ministry of Defence (WESEE) — sih.gov.in/sih2026PS

INDUSTRY REPORTS & INCIDENTS

- [11] IBM, Cost of a Data Breach Report 2025 — India average ₹22 crore, a record high.
- [12] Ponemon Institute & DTEX, 2026 Cost of Insider Risks Global Report — US\$19.5 M per organisation per year.
- [13] Verizon, 2025 Data Breach Investigations Report — human element in ~60 % of breaches.
- [14] East Asia Forum, “India’s leaky submarines,” 29 Oct 2016 — the 22,400-page Scorpène leak.



HOW WE RESEARCH: pre-registered experiments · thresholds frozen before testing · every failed arm kept in the evidence · independent spec + code review of every research branch · synthetic documents only



OPEN SOURCE & REPRODUCIBLE

github.com/ashebbbar-dev/SIH

Code, tests, benchmarks and evidence JSON: every number in this deck regenerates offline with the repo’s commands.